

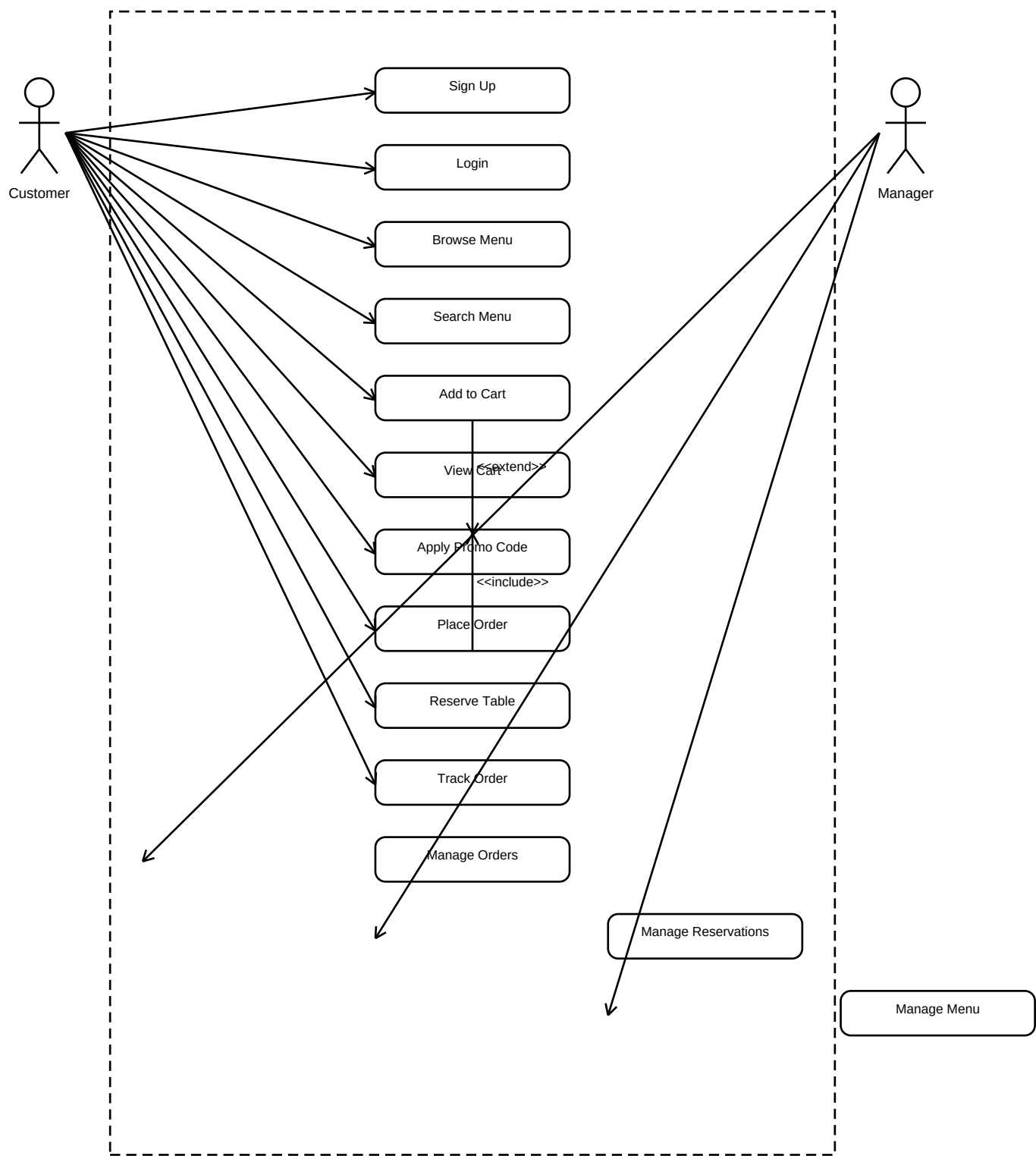
Restaurant App MVP — Complete UML Diagrams

Assignment No. 1 • Frontend Only • React Native / Expo • Fall 2026

Diagram	Assignment requirement covered
Use Case	Customer + Restaurant Manager; 12+ use cases; include/extend relationships.
Class	Required domain classes, attributes/types, operations, inheritance and multiplicities.
Sequence	Customer places an order from Add to Cart to Order Tracking.
State Machine	Pending, Preparing, Ready, Served, Cancelled with transition events.
Component	React Native screens, context providers, reducers and custom hooks.

1. Use Case Diagram

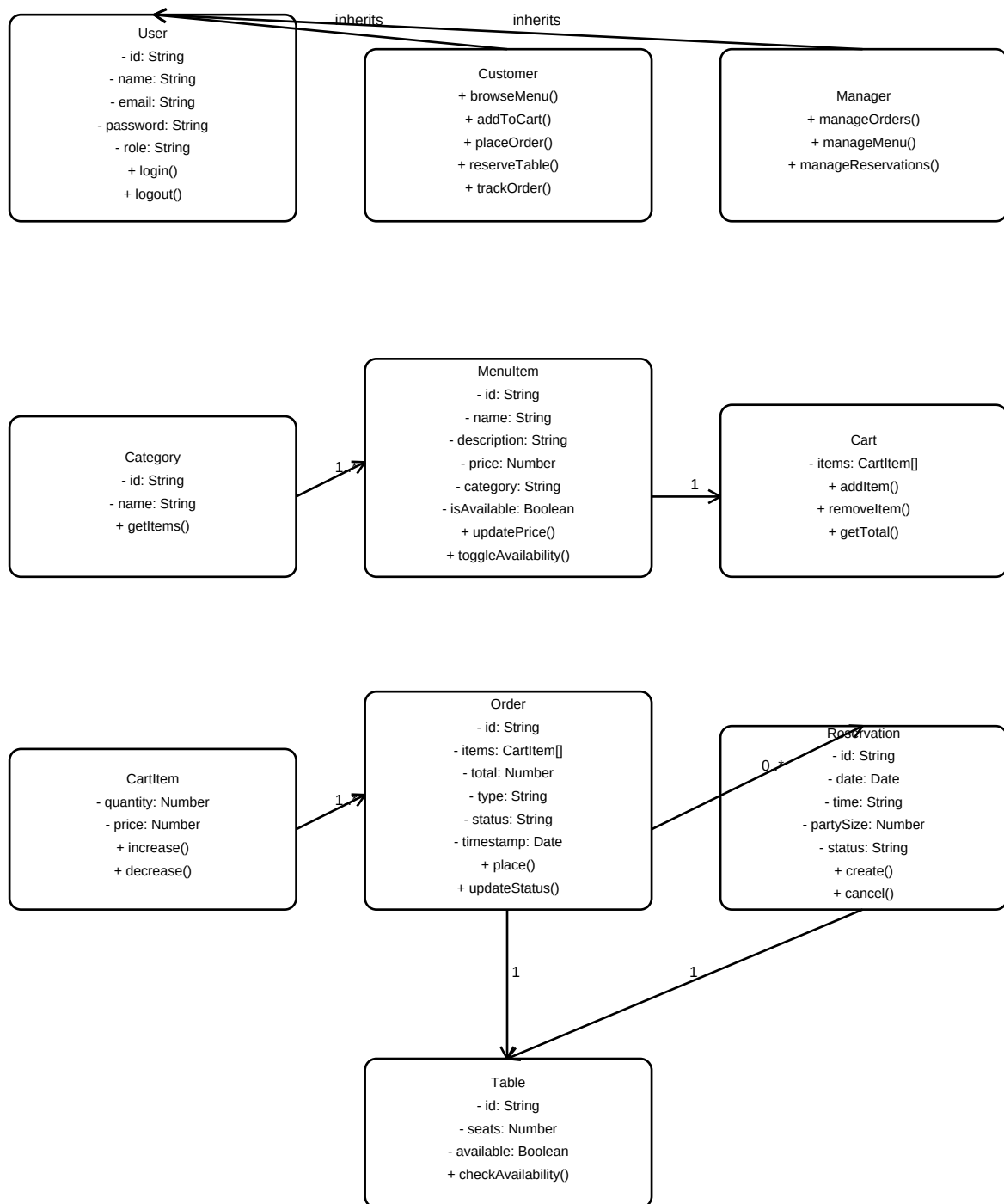
Use Case Diagram — Restaurant App MVP



Customer and Manager interactions; Place Order includes Apply Promo Code; promo application extends cart flow.

2. Class Diagram

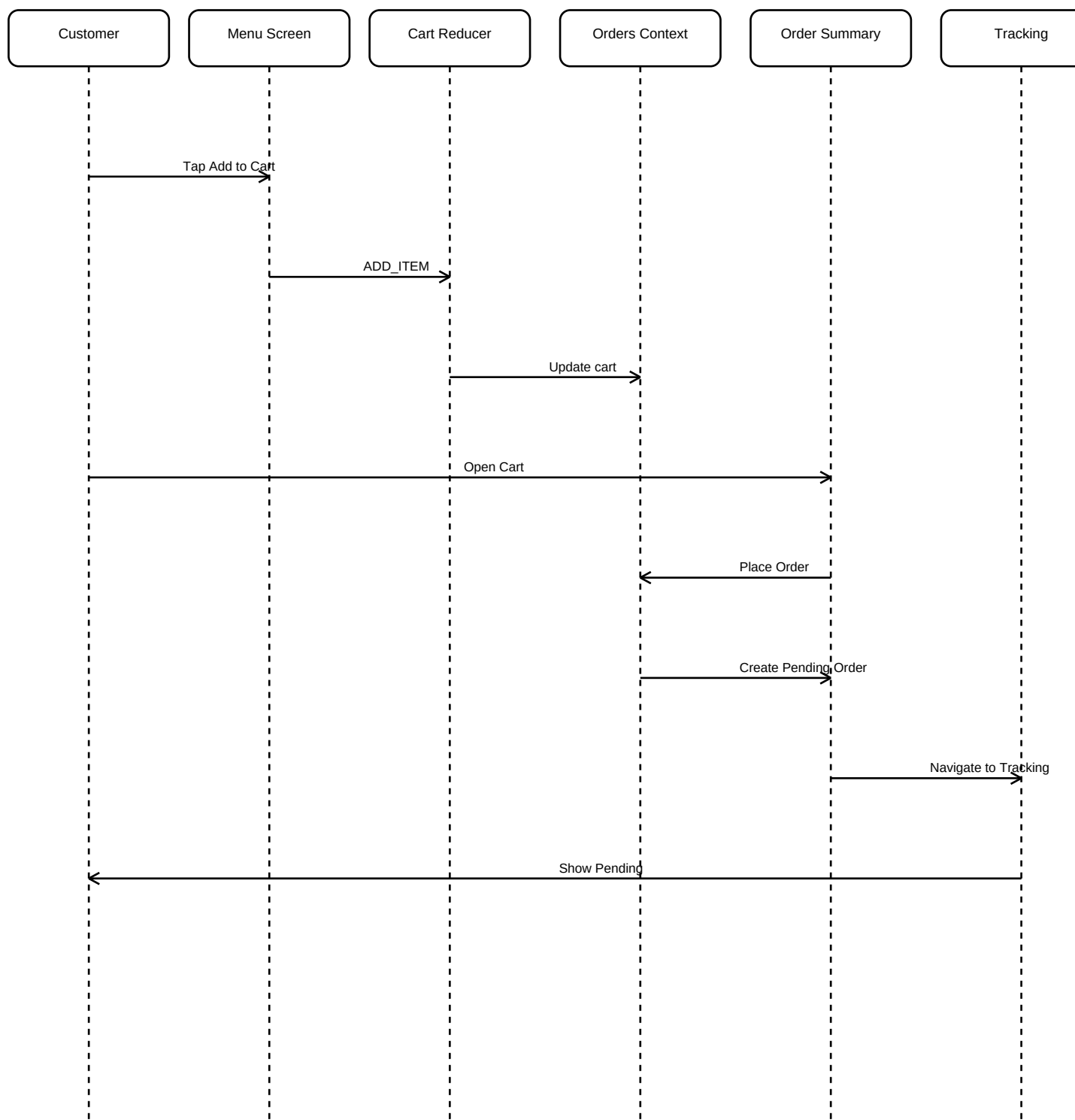
Class Diagram — Restaurant App MVP



Customer and Manager inherit from User; key associations include multiplicities.

3. Sequence Diagram

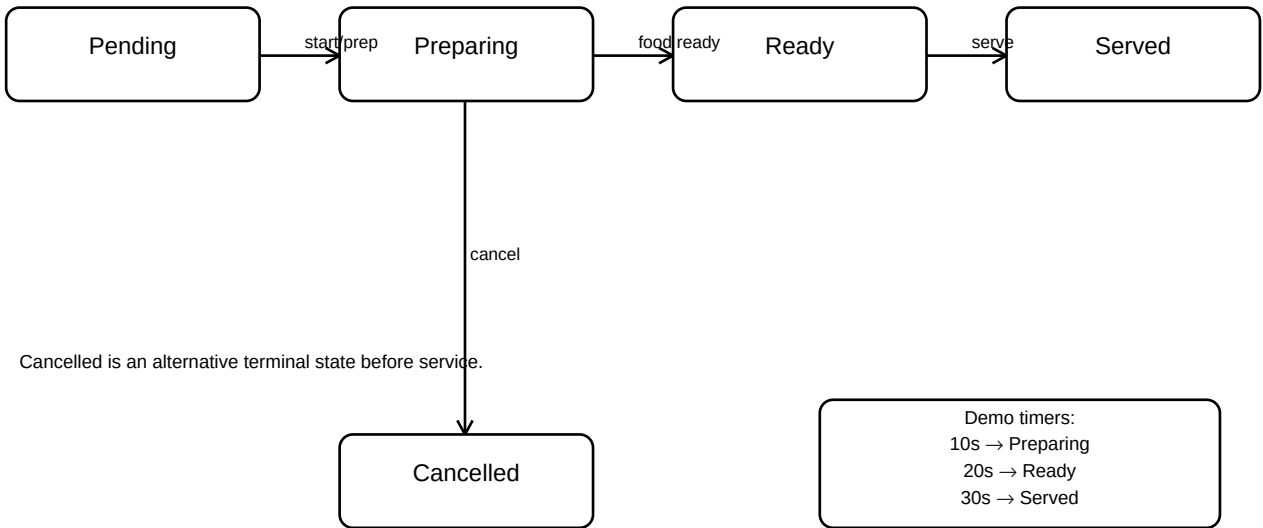
Sequence Diagram — Customer Places an Order



Scenario: Add to Cart → cart state → order creation → order summary → Order Tracking.

4. State Machine Diagram

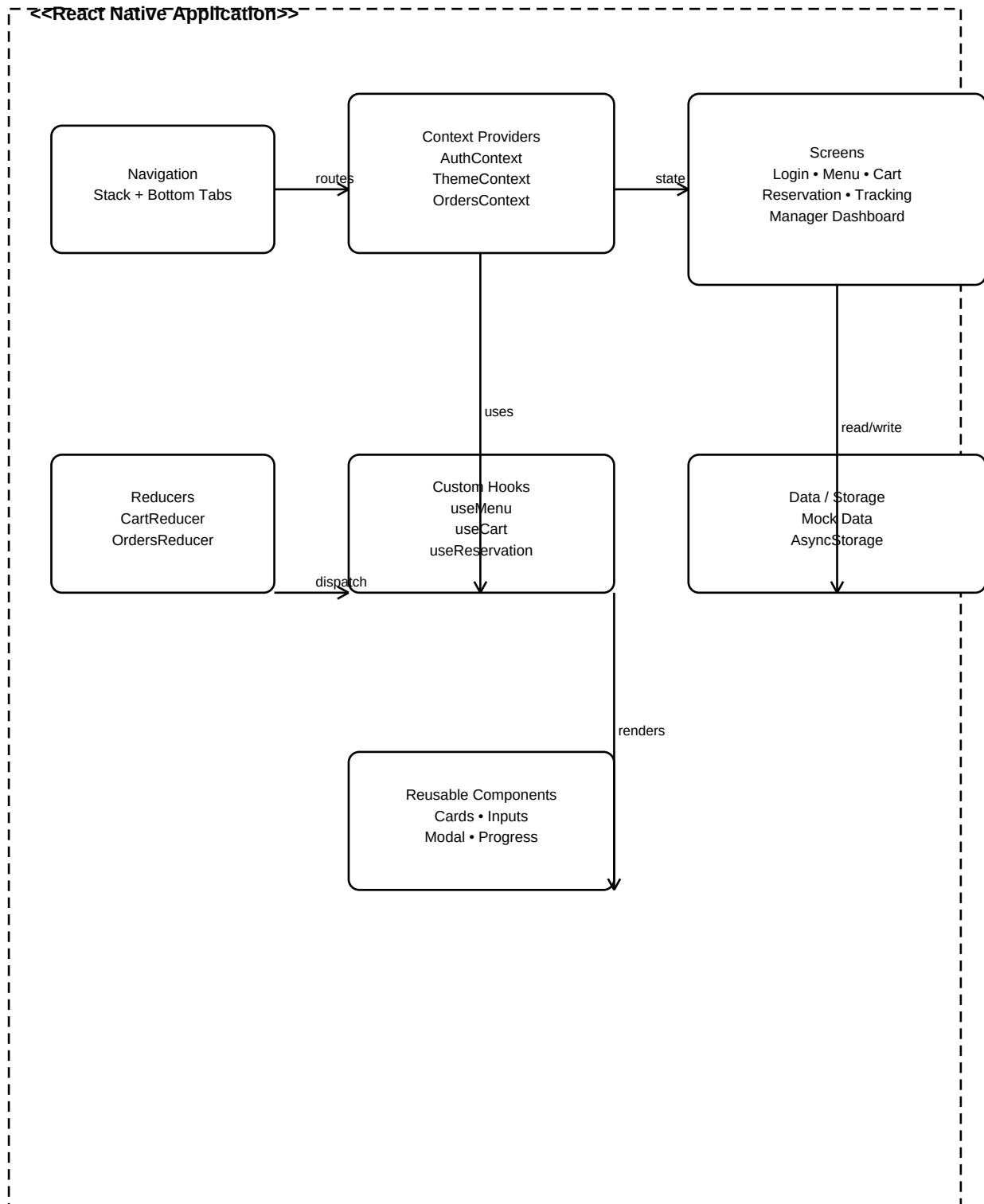
State Machine Diagram — Order Lifecycle



Required lifecycle: Pending → Preparing → Ready → Served, with Cancelled as an alternative terminal state.

5. Component Diagram

Component Diagram — React Native Restaurant App



Shows screens, context providers, reducers, custom hooks and local persistence dependencies.